INVERSION OF STABILITY FOR LOG CANONICAL PAIRS ON A FIXED GERM

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Abstract. We prove that on a fixed germ of a normal variety, if the boundary coefficients of a log canonical pair are raised to values in a fixed finite set by less than a constant depending only on the germ and the finite set, then the log canonical divisor stays $\mathbb{R}$-Cartier and the pair stays log canonical. Thus the conclusion of a conjecture of Han and Luo holds near the point of the germ for log canonical pairs, even if the germ is not klt. We also show that the conjecture fails for the threefold node, so the log canonical assumption cannot be removed.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Let $B = \sum_i b_i B_i$ be an $\mathbb{R}$-divisor on a normal variety X, where the B_i are distinct prime divisors. Following [HL23, Definition 6.1], we let

$$\|B\| := \max_i |b_i|.$$

Han and Luo proposed the following conjecture (see Definition 2.1 for klt and lc pairs $(X \ni x, B)$).

Conjecture 1.1 ([HL23, Conjecture 6.6])**.** *Let $\Gamma \subset [0,1]$ be a finite set and X a normal quasi-projective variety such that K_X is $\mathbb{R}$-Cartier. Let $x \in X$ be a closed point such that X is klt near x. Then there exists a real number $\tau > 0$ depending only on Γ and $X \ni x$ satisfying the following. Let B and B' be $\mathbb{R}$-divisors on X such that*

(1) $B' \le B$, $\|B - B'\| < \tau$, and $B \in \Gamma$, and
(2) $K_X + B'$ is $\mathbb{R}$-Cartier.

Then $K_X + B$ is $\mathbb{R}$-Cartier.

Conjecture 1.1 first appeared as [HL20, Conjecture 7.7] in the second arXiv version (2020), and then as [HL20, Conjecture 7.8] in the third arXiv version (2022). In both versions, $X \ni x$ is assumed to be a germ instead of a klt germ. In the published version [HL23] (2023), Conjecture 1.1 is asked at the end of Section 6.

Date: November 10, 2026.
2020 *Mathematics Subject Classification.* 14E30, 14J17, 14B05.
Key words and phrases. log canonical pair, inversion of stability, $\mathbb{R}$-Cartier divisor, log canonical threshold, numerically Cartier divisor.

Following [HLL25, p. 49], we call the property in Conjecture 1.1 the *inversion of stability property for $\mathbb{R}$-Cartier divisors*. The conclusion of Conjecture 1.1 was known to hold near x in the following cases:

(1) When $X \ni x$ is a terminal threefold singularity, $B' \geq 0$, and the minimal log discrepancy of $(X \ni x, B')$ is at least 1, where τ depends only on Γ [HLL25, Theorem 5.20].
(2) When $(X \ni x, B')$ is a klt pair of dimension d whose local volume [Li18] (see [HLQ24, Definition 2.1]) is at least v for a fixed real number $v > 0$, where τ depends only on d, v, and Γ; in this case, $(X \ni x, B)$ is also klt [HLQ24, Theorem 1.4].

The goal of this paper is to show that Conjecture 1.1 does not hold in general (Proposition 1.4), and that, once $(X \ni x, B')$ is assumed to be an lc pair, the conclusion of Conjecture 1.1 holds near x for any normal quasi-projective variety X and closed point $x \in X$, with no klt assumption on X. Moreover, $(X \ni x, B)$ is lc (Theorem 1.2).

The main result of this paper is the following theorem.

Theorem 1.2. *Let $\Gamma \subset [0,1]$ be a finite set, X a normal quasi-projective variety, and $x \in X$ a closed point. Then there exists a real number $\tau > 0$ depending only on Γ and $X \ni x$ satisfying the following. Let B and B' be $\mathbb{R}$-divisors on X such that*

(1) $0 \leq B' \leq B$, $\|B - B'\| < \tau$, and $B \in \Gamma$, and
(2) $(X \ni x, B')$ is an lc pair.

Then $K_X + B$ is $\mathbb{R}$-Cartier near x, and $(X \ni x, B)$ is lc.

Remark 1.3. Theorem 1.2 does not assume that X is $\mathbb{Q}$-Gorenstein or $\mathbb{Q}$-factorial, that X has rational singularities, or that (X, Δ) is klt near x for some $\mathbb{R}$-divisor Δ, and the supports of B and B' are arbitrary.

Condition (2) of Theorem 1.2 cannot be weakened to the condition that $K_X + B'$ is $\mathbb{R}$-Cartier, already when $X \ni x$ is a terminal threefold singularity and $\Gamma = \{1\}$.

Proposition 1.4. *Let $X \subset \mathbb{A}^4$ be the hypersurface $xy = zw$ and $o \in X$ the origin. Identify X with the affine cone over $Q := \mathbb{P}^1_{[s:t]} \times \mathbb{P}^1_{[u:v]}$ with respect to $\mathcal{O}_Q(1,1)$ via $x = su$, $y = tv$, $z = sv$, and $w = tu$. For any integer $n \geq 2$, let $D_{n,1}$ and $D_{n,2}$ be the cones over the curves $\{s^{n+1}u + t^{n+1}v = 0\}$ and $\{su^n + tv^n = 0\}$ on Q respectively, and let*

$$B_n := D_{n,1} + D_{n,2}, \quad B'_n := \Big(1 - \frac{1}{n}\Big) D_{n,1} + D_{n,2}.$$

Then X is terminal, K_X is Cartier, and for any integer $n \geq 2$:

(1) $D_{n,1}$ and $D_{n,2}$ are distinct prime divisors containing o,
(2) nB'_n is a principal divisor and $K_X + B'_n$ is $\mathbb{Q}$-Cartier, and
(3) $K_X + B_n$ is not $\mathbb{R}$-Cartier on any neighborhood of o.

For the node $X \ni o$ of Proposition 1.4, let $\tau > 0$ be a real number, and let $n \geq 2$ be an integer such that $1/n < \tau$. Since X is terminal and K_X is Cartier, by Proposition 1.4(1)(2) and $B_n - B'_n = \frac{1}{n} D_{n,1}$,

$$B'_n \leq B_n, \quad \|B_n - B'_n\| = \frac{1}{n} < \tau, \quad B_n \in \{1\}, \quad \text{and} \quad K_X + B'_n \text{ is } \mathbb{R}\text{-Cartier},$$

so $B := B_n$ and $B' := B'_n$ satisfy conditions (1) and (2) of Conjecture 1.1 for $X \ni o$ and $\Gamma = \{1\}$. By Proposition 1.4(3), $K_X + B_n$ is not $\mathbb{R}$-Cartier. Since τ is arbitrary, Conjecture 1.1 fails for $X \ni o$ and $\Gamma = \{1\}$. The boundaries B_n and B'_n on $X \ni o$ also disprove [HL20, Conjecture 7.7] in the second arXiv version and [HL20, Conjecture 7.8] in the third.

Sketch of the proof. We may assume that $\Gamma \not\subset \{0\}$, and let $d := \dim X$ and $\gamma := \min(\Gamma \backslash \{0\})$. Possibly shrinking X to a neighborhood of x, we may assume that (X, Δ) is an lc pair for some $\mathbb{R}$-divisor Δ, as otherwise no B' satisfies condition (2) of Theorem 1.2. There exists a projective

birational morphism $f : Y \to X$ from a $\mathbb{Q}$-factorial variety Y such that, letting E be the reduced f-exceptional divisor, any lc pair $(X \ni x, B')$ satisfies

$$K_Y + E + f_*^{-1}B' = f^*(K_X + B') \quad \text{and} \quad (Y, E + f_*^{-1}B') \text{ is lc}$$

over a neighborhood of x (see Construction-Proposition 4.2). In the proof of Theorem 1.2 in Section 5, **Steps 1–3** fix f and τ in terms of $X \ni x$ and Γ, before B and B' are given. Let $B_Y := f_*^{-1}B$. As Y is $\mathbb{Q}$-factorial, B_Y and $K_Y + E + B_Y$ are $\mathbb{R}$-Cartier. Since $f_*(E + B_Y) = B$, it suffices to show that, over a neighborhood of x, $(Y, E + B_Y)$ is lc and $K_Y + E + B_Y$ is numerically trivial over X, see Lemma 2.4 (cf. [Has20, Theorem 5.5]) and **Step 8**. The pair $(Y, E + B_Y)$ is lc by the ACC for lc thresholds [HMX14, Theorem 1.1] (Theorem 2.3): **Step 1** fixes a real number $\epsilon \in (0, 1)$ such that

$$\mathrm{LCT}_d(\{0, 1\}, \Gamma\backslash\{0\}) \cap (1 - \epsilon, 1) = \emptyset,$$

and **Step 3** takes $\tau \leq \gamma\epsilon/2$, so $f_*^{-1}B' \geq (1 - \epsilon/2)B_Y$. Thus $\mathrm{lct}(Y, E; B_Y) \geq 1 - \epsilon/2$, and hence $\mathrm{lct}(Y, E; B_Y) \geq 1$ (**Step 7**).

Steps 4–6 prove that $K_Y + E + B_Y$ is numerically trivial over X near x. **Step 2** fixes a projective birational morphism $q : Y' \to Y$ from a smooth variety Y' such that $F := (f \circ q)^{-1}(x)$, with the reduced structure, is an snc divisor. Since $(X \ni x, B')$ is lc, intersecting $(f \circ q)^*(K_X + B')$ with curves on F bounds the number of components B_i of B and the degrees of $(f \circ q)_*^{-1}B_i|_{F_k}$ for the components F_k of F, so the numerical classes of $(f \circ q)_*^{-1}B_i|_{F_k}$ belong to a finite set (**Step 4**). Thus the class z_B of $K_{Y'} + q_*^{-1}E + (f \circ q)_*^{-1}B$ on the components of F, modulo the q-exceptional prime divisors, belongs to a finite set $\mathcal{S}$ fixed in **Step 3**. By the choice of τ in **Step 3**, z_B is closer to the class for B', which is 0, than any non-zero element of $\mathcal{S}$, so $z_B = 0$ (**Step 5**). Lemma 2.5 and the negativity lemma conclude (**Step 6**). In Section 4, $f = h \circ g$, where h and g are given in Lemma 4.1 and [HMX14, Proposition 3.3.1], respectively.

Structure of the paper. Section 2 recalls the ACC for log canonical thresholds, shows that $K_X + B$ is $\mathbb{R}$-Cartier and (X, B) is lc if (Y, B_Y) is an lc pair on a higher model $f : Y \to X$ such that $f_*B_Y = B$ and $K_Y + B_Y$ is numerically trivial over X, and shows that numerical triviality on the closed fiber over a closed point $x \in X$ spreads to a neighborhood of x. In Section 3, we study the boundaries $B'_n \leq B_n$ on the threefold node $xy = zw$. In Section 4, we construct a higher model of X that does not depend on the lc pair $(X \ni x, B')$. In Section 5, we prove the main theorem.

Acknowledgements. The work was partially supported by the National Key R&D Program of China #2024YFA1014400. Jihao Liu would like to thank Bin Dong, Guoxiong Gao, Zeming Sun, Bin Wu, and Leheng Chen for AI assistance. Jihao Liu would like to thank Ruochuan Liu and Gang Tian for consistent support and encouragement.

2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs.

Definition 2.1. A *pair* $(X/Z \ni z, B)$ consists of a contraction $\pi : X \to Z$ between normal quasi-projective varieties, a point $z \in Z$, and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier over a neighborhood of z. If π is the identity morphism, then we may drop Z and say that $(X \ni z, B)$ is a pair. If $(X/Z \ni z, B)$ is a pair for any $z \in Z$, then we say that $(X/Z, B)$ is a pair and also say that (X, B) is a pair. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \mathrm{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B).

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X. We say that a pair $(X \ni x, B)$ is *lc* (resp. *klt*) if $(U, B|_U)$ is lc (resp. klt) for some open neighborhood U of x.

Lemma 2.2. *Let X be a normal variety, $D \geq 0$ an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on X, and $h : X' \to X$ a birational morphism from a normal variety X'. Then $h^*D \geq 0$.*

Proof. The question is local on X, so we may assume that $D = \sum_{i=1}^m r_i D_i$, where $r_1, \ldots, r_m \in \mathbb{R}$ and $D_1, \ldots, D_m$ are Cartier divisors on X. The coefficients of $\sum_{i=1}^m t_i D_i$ along the finitely many prime divisors in $\bigcup_{i=1}^m \operatorname{Supp} D_i$ are linear forms in $(t_1, \ldots, t_m)$ with integer coefficients, so

$$\mathcal{C} := \Big\{(t_1, \ldots, t_m) \in \mathbb{R}^m \;\Big|\; \sum_{i=1}^m t_i D_i \geq 0\Big\}$$

is a rational polyhedral cone. Thus there exist $(t_{k,1}, \ldots, t_{k,m}) \in \mathcal{C} \cap \mathbb{Q}^m$ for $k \geq 1$ such that

$$(r_1, \ldots, r_m) = \lim_{k \to +\infty} (t_{k,1}, \ldots, t_{k,m}).$$

For any k, a positive integral multiple of $\sum_{i=1}^m t_{k,i} D_i$ is a Cartier divisor ≥ 0, so

$$h^*\Big(\sum_{i=1}^m t_{k,i} D_i\Big) \geq 0.$$

For any prime divisor P on X',

$$\operatorname{mult}_P h^*D = \sum_{i=1}^m r_i \operatorname{mult}_P h^*D_i = \lim_{k \to +\infty} \sum_{i=1}^m t_{k,i} \operatorname{mult}_P h^*D_i \geq 0.$$

The lemma follows. □

2.2. **Log canonical thresholds.**

Theorem 2.3 ([HMX14, Theorem 1.1])**.** *Let d be a positive integer, and let $\Gamma_1 \subset [0, 1]$ and $\Gamma_2 \subset (0, +\infty)$ be DCC sets. For any lc pair (X, B) and any $\mathbb{R}$-Cartier $\mathbb{R}$-divisor $D \geq 0$ on X, let*

$$\operatorname{lct}(X, B; D) := \sup\{t \in \mathbb{R} \mid (X, B + tD) \text{ is lc}\}.$$

Let $\operatorname{LCT}_d(\Gamma_1, \Gamma_2)$ be the set of $\operatorname{lct}(X, B; D)$, where (X, B) is an lc pair of dimension d such that $B \in \Gamma_1$, and $D \geq 0$ is an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on X such that $D \in \Gamma_2$. Then $\operatorname{LCT}_d(\Gamma_1, \Gamma_2)$ satisfies the ACC.

2.3. **Descent of lc and $\mathbb{R}$-Cartier from a higher model.**

Lemma 2.4 (cf. [Has20, Theorem 5.5])**.** *Let X be a normal quasi-projective variety, $B \geq 0$ an $\mathbb{R}$-divisor on X, and $f : Y \to X$ a projective birational morphism. Let B_Y be an $\mathbb{R}$-divisor on Y such that*

(1) $f_*B_Y = B$,
(2) (Y, B_Y) is an lc pair, and
(3) $K_Y + B_Y$ is numerically trivial over X.

Then $K_X + B$ is $\mathbb{R}$-Cartier and (X, B) is lc.

Proof. Following [Has20, Definition 5.3], an $\mathbb{R}$-divisor D on X is *numerically Cartier* if there exist a resolution $g : \widetilde{X} \to X$ and an $\mathbb{R}$-divisor $\widetilde{D}$ on $\widetilde{X}$ such that $\widetilde{D}$ is numerically trivial over X and $g_*\widetilde{D} = D$.

We use the following notions of [BdFF12], all over X and with real coefficients: Weil $\boldsymbol{b}$-divisors W, their traces $W_{\widetilde{X}}$ on higher models $\widetilde{X} \to X$, and Cartier $\boldsymbol{b}$-divisors are as in [BdFF12, Section 1.2], and $\overline{\widetilde{D}}$ denotes the Cartier $\boldsymbol{b}$-divisor determined by an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor $\widetilde{D}$ on a

higher model. The X-nef Cartier and Weil $\boldsymbol{b}$-divisors are as in [BdFF12, Section 2.2, Definition 2.9]. The nef envelope $\mathrm{Env}_X(D)$ of an $\mathbb{R}$-divisor D on X is as in [BdFF12, Definition 2.3], and by [BdFF12, Corollary 2.13], $\mathrm{Env}_X(D)$ is the largest X-nef Weil $\boldsymbol{b}$-divisor W such that $W_X \le D$, and $\mathrm{Env}_X(0) = 0$.

By [Has20, Theorem 5.5], since X is a normal quasi-projective variety and $B \ge 0$, it suffices to show that (i) $K_X + B$ is numerically Cartier, and (ii) for any log resolution $h : X' \to X$ of (X, B), the coefficient of any prime divisor P on X' in $K_{X'} + \mathrm{Env}_X(-(K_X + B))_{X'}$ is ≥ -1.

Step 1. Let $\rho : Y' \to Y$ be a log resolution of (Y, B_Y), and let

$$L := \rho^*(K_Y + B_Y).$$

By the projection formula, L is numerically trivial over X. Moreover,

$$(f \circ \rho)_* L = f_*(K_Y + B_Y) = K_X + B.$$

Since Y' is smooth and $f \circ \rho : Y' \to X$ is a projective birational morphism, $f \circ \rho$ is a resolution of X. Thus (i) holds.

Step 2. We show that

$$\mathrm{Env}_X(-(K_X + B)) = -\overline{L}.$$

Since L is numerically trivial over X, both $\overline{L}$ and $-\overline{L}$ are X-nef. Since

$$(-\overline{L})_X = -(f \circ \rho)_* L = -(K_X + B),$$

[BdFF12, Corollary 2.13] implies that $-\overline{L} \le \mathrm{Env}_X(-(K_X + B))$. Conversely, let

$$W_0 := \mathrm{Env}_X(-(K_X + B)) + \overline{L}.$$

Then W_0 is X-nef, as the X-nef Weil $\boldsymbol{b}$-divisors form a convex cone [BdFF12, Definition 2.9], and

$$(W_0)_X = \mathrm{Env}_X(-(K_X + B))_X + K_X + B \le -(K_X + B) + K_X + B = 0.$$

By [BdFF12, Corollary 2.13], $W_0 \le \mathrm{Env}_X(0) = 0$, that is, $\mathrm{Env}_X(-(K_X + B)) \le -\overline{L}$.

Step 3. Let $h : X' \to X$ be a log resolution of (X, B) and P a prime divisor on X'. Let X'' be a smooth variety with projective birational morphisms $\phi : X'' \to X'$ and $\psi : X'' \to Y'$ such that $h \circ \phi = f \circ \rho \circ \psi$. By **Step 2**,

$$K_{X'} + \mathrm{Env}_X(-(K_X + B))_{X'} = K_{X'} - \phi_*\psi^* L = \phi_*\big(K_{X''} - (\rho \circ \psi)^*(K_Y + B_Y)\big).$$

Thus the coefficient of P in $K_{X'} + \mathrm{Env}_X(-(K_X + B))_{X'}$ is the coefficient of $\phi_*^{-1}P$ in

$$K_{X''} - (\rho \circ \psi)^*(K_Y + B_Y),$$

which is

$$a(\phi_*^{-1}P, Y, B_Y) - 1 \ge -1,$$

since (Y, B_Y) is lc. Thus (ii) holds.

The lemma follows. □

2.4. **Numerical triviality near a fiber.**

Lemma 2.5. *Let Y be a smooth variety, $f : Y \to X$ a projective morphism to a variety X, $x \in X$ a closed point, and L an $\mathbb{R}$-divisor on Y such that $L \cdot C = 0$ for any curve $C \subset f^{-1}(x)$. Then L is numerically trivial over a neighborhood of x.*

Proof. Let F be the reduced closed subscheme of Y with support $f^{-1}(x)$. Then F is a possibly reducible projective variety.

Step 1. Since Y is smooth, $L = \sum_{i=1}^m r_i L_i$ for some $r_1, \dots, r_m \in \mathbb{R}$ and Cartier divisors $L_1, \dots, L_m$ on Y. Let

$$V := \Big\{(t_1, \dots, t_m) \in \mathbb{R}^m \;\Big|\; \sum_{i=1}^m t_i(L_i \cdot C) = 0 \text{ for any curve } C \subset F\Big\}.$$

Since $L_i \cdot C \in \mathbb{Z}$ for any i and any C, V is the solution space of a system of homogeneous linear equations with integer coefficients, so V is spanned by finitely many vectors $v_1, \dots, v_n \in$

$\mathbb{Z}^m$, where $v_j = (v_{j,1}, \dots, v_{j,m})$. Since $(r_1, \dots, r_m) \in V$, there exist $s_1, \dots, s_n \in \mathbb{R}$ such that $(r_1, \dots, r_m) = \sum_{j=1}^n s_j v_j$. Let $M_j := \sum_{i=1}^m v_{j,i} L_i$ for any j. Then $M_1, \dots, M_n$ are Cartier divisors on Y,

$$L = \sum_{j=1}^n s_j M_j, \quad \text{and} \quad M_j \cdot C = 0 \text{ for any } j \text{ and any curve } C \subset F.$$

Step 2. By [BdFFU15, Lemma 5.13], if T is a possibly reducible projective variety, then a line bundle $\mathcal{M}$ on T satisfies $\mathcal{M} \cdot C = 0$ for any curve $C \subset T$ if and only if $c_1(\mathcal{M}) = 0 \in H^2(T, \mathbb{Q})$. By **Step 1**,

$$c_1(\mathcal{O}_Y(M_j)|_F) = 0 \in H^2(F, \mathbb{Q}) \quad \text{for any } j.$$

Step 3. Let $\mathbb{Q}_Y$ be the constant sheaf on Y^{an}. For any j, let $\sigma_j \in H^0(X^{an}, R^2 f_* \mathbb{Q}_Y)$ be the image of $c_1(\mathcal{O}_Y(M_j)) \in H^2(Y^{an}, \mathbb{Q})$ under the natural map

$$H^2(Y^{an}, \mathbb{Q}) \to H^0(X^{an}, R^2 f_* \mathbb{Q}_Y).$$

By proper base change [KS90, Proposition 2.6.7], for any $x' \in X$, $(R^2 f_* \mathbb{Q}_Y)_{x'} \cong H^2(f^{-1}(x'), \mathbb{Q})$, and under the isomorphism the germ $(\sigma_j)_{x'}$ of σ_j at x' is $c_1(\mathcal{O}_Y(M_j)|_{f^{-1}(x')})$. By **Step 2**, $(\sigma_j)_x = 0$ for any j.

Step 4. For any j, let

$$\Sigma_j := \{x' \in X \mid (\sigma_j)_{x'} \neq 0\}.$$

Then Σ_j is closed in X^{an} and $x \notin \Sigma_j$. By [Dim04, Theorem 4.1.5], there exists a finite stratification $X = \bigsqcup_{k=1}^N X_k$ such that for any k, X_k is a connected locally closed algebraic subset of X and $R^2 f_* \mathbb{Q}_Y|_{X_k}$ is a local system. For any j and k, $\sigma_j|_{X_k}$ is a section of a local system on the connected set X_k, so $\Sigma_j \cap X_k$ is open and closed in X_k, that is, $\Sigma_j \cap X_k \in \{\emptyset, X_k\}$. Thus Σ_j is a union of strata, hence an algebraically constructible subset of X. By [Mum99, Chapter I, Section 10, Corollary 1], the Zariski closure of Σ_j is equal to the closure of Σ_j in X^{an}, which is Σ_j. Therefore Σ_j is Zariski closed, and

$$X^\circ := X \setminus \bigcup_{j=1}^n \Sigma_j$$

is a Zariski open neighborhood of x.

Let $x' \in X^\circ$ and let $C \subset f^{-1}(x')$ be a curve. For any j, $c_1(\mathcal{O}_Y(M_j)|_{f^{-1}(x')}) = (\sigma_j)_{x'} = 0$, so $M_j \cdot C = 0$. Thus

$$L \cdot C = \sum_{j=1}^n s_j (M_j \cdot C) = 0.$$

Hence L is numerically trivial over X°. The lemma follows. □

3. Necessity of the lc assumption

In this section we prove Proposition 1.4.

Proof of Proposition 1.4. **Step 1.** In this step we prove that X is terminal and K_X is Cartier.

The partial derivatives of $xy - zw$ are y, x, $-w$, $-z$, so $\operatorname{Sing} X = \{o\}$ and $\operatorname{codim}_X \{o\} = 3$. Since X is a hypersurface in $\mathbb{A}^4$ and $\operatorname{codim}_X \operatorname{Sing} X = 3$, X is normal, and $K_X \sim (K_{\mathbb{A}^4} + X)|_X$ is Cartier.

Let $\pi : Y \to X$ be the blow-up of o and F the exceptional prime divisor. Y is covered by 4 affine charts, each isomorphic to $\mathbb{A}^3$. For instance, on the chart where $\pi^* x$ is a local equation of F, the functions x, z/x, w/x are coordinates and

$$y = x \cdot \frac{z}{x} \cdot \frac{w}{x};$$

the charts of y, z, w are obtained by exchanging $x \leftrightarrow y$, $(x, y) \leftrightarrow (z, w)$, and both. In particular, Y is smooth. Moreover, Y is the total space of the line bundle $\mathcal{O}_Q(-1, -1)$ and F is the zero

section. Let $g : Y \to Q$ be the bundle projection. Over $Y\backslash F \cong X\backslash\{o\}$, g sends a point of $X\backslash\{o\}$ to the corresponding point $([s : t], [u : v])$ of Q. On the chart above, $g = ([1 : w/x], [1 : z/x])$. Thus $g|_F : F \to Q$ is an isomorphism and

$$\mathcal{O}_Y(F)|_F \cong (g|_F)^*\mathcal{O}_Q(-1, -1).$$

Let $h : W \to \mathbb{A}^4$ be the blow-up of the origin and G the exceptional prime divisor. Then Y is the strict transform of X, $h^*X = Y + 2G$ as $\mathrm{mult}_o X = 2$, and $G|_Y = F$ (on the chart above, both are defined by x). Since $K_W = h^*K_{\mathbb{A}^4} + 3G$, adjunction implies

$$K_Y = (K_W + Y)|_Y = \big(h^*(K_{\mathbb{A}^4} + X) + 3G - 2G\big)|_Y = \pi^*K_X + F. \tag{3.1}$$

Since Y is smooth, $\mathrm{Exc}(\pi) = F$ is a smooth prime divisor, and $a(F, X, 0) = 2$ by (3.1), X is terminal.

Step 2. In this step we prove (1).

Let

$$C_{n,1} := \{s^{n+1}u + t^{n+1}v = 0\} \subset Q, \quad C_{n,2} := \{su^n + tv^n = 0\} \subset Q.$$

Since $s^{n+1}u + t^{n+1}v$ is linear in u, v and s^{n+1}, t^{n+1} have no common zero on $\mathbb{P}^1_{[s:t]}$, the zero scheme $C_{n,1}$ is the graph of the morphism $\mathbb{P}^1_{[s:t]} \to \mathbb{P}^1_{[u:v]}$ given by $[u : v] = [-t^{n+1} : s^{n+1}]$. Similarly, $C_{n,2}$ is the graph of the morphism $\mathbb{P}^1_{[u:v]} \to \mathbb{P}^1_{[s:t]}$ given by $[s : t] = [-v^n : u^n]$. Thus $C_{n,1} \cong C_{n,2} \cong \mathbb{P}^1$ are smooth irreducible curves,

$$\mathcal{O}_Q(C_{n,1}) \cong \mathcal{O}_Q(n + 1, 1), \quad \mathcal{O}_Q(C_{n,2}) \cong \mathcal{O}_Q(1, n),$$

and $C_{n,1} \neq C_{n,2}$ since $(n + 1, 1) \neq (1, n)$ for $n \geq 2$.

By the description of $g|_{Y\backslash F}$ in **Step 1**, $D_{n,1} = \pi(g^{-1}(C_{n,1}))$ and $D_{n,2} = \pi(g^{-1}(C_{n,2}))$. The surfaces $g^{-1}(C_{n,1})$ and $g^{-1}(C_{n,2})$ are line bundles over $C_{n,1}$ and $C_{n,2}$ respectively, hence irreducible, distinct, not contained in F, and meeting F. Since $\pi(F) = o$, $D_{n,1}$ and $D_{n,2}$ are distinct prime divisors containing o, which is (1). We let

$$\widetilde{D}_{n,1} := \pi_*^{-1}D_{n,1} = g^*C_{n,1}, \quad \widetilde{D}_{n,2} := \pi_*^{-1}D_{n,2} = g^*C_{n,2},$$

so

$$\mathcal{O}_Y(\widetilde{D}_{n,1})|_F \cong (g|_F)^*\mathcal{O}_Q(n + 1, 1), \quad \mathcal{O}_Y(\widetilde{D}_{n,2})|_F \cong (g|_F)^*\mathcal{O}_Q(1, n). \tag{3.2}$$

Step 3. In this step we prove (2).

Let $N_n := n^2 + n - 1$ and

$$\sigma_n := (s^{n+1}u + t^{n+1}v)^{n-1}(su^n + tv^n)^n.$$

The bidegree of σ_n is (N_n, N_n):

$$(n - 1)(n + 1) + n = N_n,$$
$$(n - 1) \cdot 1 + n \cdot n = N_n.$$

Any monomial in s, t, u, v of bidegree (N_n, N_n) is a product of N_n monomials among su, tv, sv, tu, so σ_n is a homogeneous polynomial of degree N_n in x, y, z, w. Let φ_n be the regular function on X defined by σ_n. On the chart of **Step 1**,

$$\pi^*\varphi_n = x^{N_n} \cdot \sigma_n\Big(1, \frac{w}{x}, 1, \frac{z}{x}\Big),$$

where the second factor is the pullback under g of σ_n on the affine chart $\{su \neq 0\}$ of Q with coordinates t/s, v/u. Since x is a unit on the chart of **Step 1** minus F, and similarly on the other three charts, and $\mathrm{div}(\sigma_n) = (n - 1)C_{n,1} + nC_{n,2}$ by **Step 2**,

$$\mathrm{div}(\pi^*\varphi_n)|_{Y\backslash F} = \big((n - 1)\widetilde{D}_{n,1} + n\widetilde{D}_{n,2}\big)|_{Y\backslash F}.$$

Since π induces $Y\backslash F \cong X\backslash\{o\}$ and $\mathrm{codim}_X\{o\} = 3$,

$$\mathrm{div}_X(\varphi_n) = (n - 1)D_{n,1} + nD_{n,2} = nB'_n.$$

Thus nB'_n is a principal divisor. Since K_X is Cartier, $K_X + B'_n$ is $\mathbb{Q}$-Cartier, which is (2).

Step 4. In this step we prove (3).

Let U be an open neighborhood of o, and assume that B_n is $\mathbb{R}$-Cartier on U. Possibly replacing U with a smaller open neighborhood of o, we may write

$$B_n = \sum_{i=1}^{m} r_i \operatorname{div}(\psi_i)$$

on U, where $r_1, \dots, r_m$ are real numbers and $\psi_1, \dots, \psi_m$ are rational functions on X. Since π is an isomorphism over $U\backslash\{o\}$,

$$\pi^* B_n := \sum_{i=1}^{m} r_i \operatorname{div}(\pi^*\psi_i) = \widetilde{D}_{n,1} + \widetilde{D}_{n,2} + cF$$

on $\pi^{-1}(U)$ for some real number c. The map $D \to \mathcal{O}_Y(D)|_F$ induces an $\mathbb{R}$-linear map from $\mathbb{R}$-divisors on $\pi^{-1}(U)$ to $\operatorname{Pic}(F) \otimes \mathbb{R}$ which vanishes on $\operatorname{div}(\pi^*\psi_i)$ for any i. Since

$$\operatorname{Pic}(Q) \otimes \mathbb{R} = \mathbb{R}[\mathcal{O}_Q(1,0)] \oplus \mathbb{R}[\mathcal{O}_Q(0,1)],$$

(3.2) and $\mathcal{O}_Y(F)|_F \cong (g|_F)^* \mathcal{O}_Q(-1,-1)$ imply

$$n + 2 - c = 0 \quad \text{and} \quad n + 1 - c = 0,$$

a contradiction. Thus B_n is not $\mathbb{R}$-Cartier on any neighborhood of o. Since K_X is Cartier, $K_X + B_n$ is not $\mathbb{R}$-Cartier on any neighborhood of o, which is (3). □

4. A higher model independent of the boundary

In this section we prove Construction-Proposition 4.2.

Lemma 4.1 (cf. [Has20, Theorem 4.17])**.** *Let X be a normal quasi-projective variety such that (X, B) is an lc pair for some $\mathbb{R}$-divisor B on X. Then there exists a small birational morphism $h : X' \to X$ such that $K_{X'}$ is $\mathbb{R}$-Cartier and $(X', 0)$ is lc.*

Proof. We recall [Has20, Definitions 4.1 and 4.2] for the zero boundary, written with log discrepancies. Discrepancies in [Has20] are the log discrepancies minus 1. For any prime divisor P over X, let

$$\alpha(P) := \sup\big\{a(P, U, B_U) \;\big|\; U \subset X \text{ affine open},\ U \cap \operatorname{center}_X P \neq \emptyset,\ (U, B_U) \text{ is a pair}\big\}.$$

For an $\mathbb{R}$-divisor $\Delta \geq 0$ on X, $\langle X, \Delta\rangle$ denotes X with Δ, where $K_X + \Delta$ need not be $\mathbb{R}$-Cartier [Has20]. Then $\alpha(P) - 1$ is the number $\alpha(P, X, 0)$ of [Has20, Definition 4.1], and $\langle X, 0\rangle$ is *pseudo-lc* [Has20, Definition 4.2] if

$$\alpha(P) \geq 0 \quad \text{for any prime divisor } P \text{ over } X.$$

For any prime divisor P over X and any affine open subset $U \subset X$ with $U \cap \operatorname{center}_X P \neq \emptyset$, $(U, B|_U)$ is an lc pair, so

$$\alpha(P) \geq a(P, U, B|_U) \geq 0.$$

Thus $\langle X, 0\rangle$ is pseudo-lc.

The last sentence of [Has20, Theorem 4.17] says that if $\langle X, \Delta\rangle$ is pseudo-lc for a boundary $\mathbb{R}$-divisor Δ on X, then $\langle X, \Delta\rangle$ has a small lc modification. By [Has20, Definition 2.3], an lc modification of $\langle X, \Delta\rangle$ is a projective birational morphism $h : X' \to X$ from a normal variety X' such that, letting Δ' be $h_*^{-1}\Delta$ plus the sum of all h-exceptional prime divisors with coefficient 1, $K_{X'} + \Delta'$ is an h-ample $\mathbb{R}$-Cartier $\mathbb{R}$-divisor and (X', Δ') is lc, and the lc modification is small if h is small. We take $\Delta = 0$. Since h is small, there does not exist any h-exceptional prime divisor, so $\Delta' = 0$. Thus $K_{X'}$ is $\mathbb{R}$-Cartier and $(X', 0)$ is lc. The lemma follows. □

Construction-Proposition 4.2. Let X be a normal quasi-projective variety such that (X, Δ) is an lc pair for some $\mathbb{R}$-divisor Δ on X. Then there exists a projective birational morphism $f : Y \to X$ from a $\mathbb{Q}$-factorial variety Y satisfying the following. Let E be the reduced f-exceptional divisor. Let $x \in X$ be a (not necessarily closed) point and B an $\mathbb{R}$-divisor on X such that $(X \ni x, B)$ is an lc pair, and let $B_Y := f_*^{-1}B$. Then there exists a neighborhood U of x such that, over U,

$$K_Y + E + B_Y = f^*(K_X + B)$$

and $(Y, E + B_Y)$ is lc.

Proof. By Lemma 4.1, there exists a small birational morphism $h : X' \to X$ such that $K_{X'}$ is $\mathbb{R}$-Cartier and $(X', 0)$ is lc. Since h is projective, X' is quasi-projective, so $(X', 0)$ is an lc pair. By [HMX14, Proposition 3.3.1], there exists a projective birational morphism $g : Y \to X'$ satisfying the following, where E is the reduced g-exceptional divisor:

(1) Y is $\mathbb{Q}$-factorial, and
(2) $K_Y + E = g^*K_{X'}$.

In (2), the correction term $\sum_{a(E_i,X',0)<0} a(E_i, X', 0)E_i$ of [HMX14, Proposition 3.3.1(3)], the sum over the components E_i of E, is 0 since $(X', 0)$ is lc. By (2), for any component E_i of E,

$$a(E_i, X', 0) = 1 - \operatorname{mult}_{E_i} E = 0.$$

Let $f := h \circ g$. Then f is a projective birational morphism. Since h is small, a prime divisor on Y is f-exceptional if and only if it is g-exceptional, so E is the reduced f-exceptional divisor.

Let x and B be as in the statement. Possibly shrinking X to a neighborhood of x, we may assume that (X, B) is an lc pair. Let $B_{X'} := h_*^{-1}B$. Since h is small, h_* is injective on $\mathbb{R}$-divisors on X', and

$$h_*(K_{X'} + B_{X'}) = K_X + B = h_*h^*(K_X + B),$$

so

$$K_{X'} + B_{X'} = h^*(K_X + B).$$

Thus $K_{X'} + B_{X'}$ is $\mathbb{R}$-Cartier, and since $K_{X'}$ is $\mathbb{R}$-Cartier, $B_{X'}$ is $\mathbb{R}$-Cartier. Thus $(X', B_{X'})$ is an lc pair.

Let E_i be any component of E. Since $B_{X'} \geq 0$ is $\mathbb{R}$-Cartier, by Lemma 2.2, $g^*B_{X'} \geq 0$, so $\operatorname{mult}_{E_i} g^*B_{X'} \geq 0$. By (2), $K_Y + E + g^*B_{X'} = g^*(K_{X'} + B_{X'})$, so

$$0 \leq a(E_i, X', B_{X'}) = a(E_i, X', 0) - \operatorname{mult}_{E_i} g^*B_{X'} = -\operatorname{mult}_{E_i} g^*B_{X'}.$$

Thus $\operatorname{mult}_{E_i} g^*B_{X'} = 0$ for any component E_i of E. Since $g_*^{-1}B_{X'} = f_*^{-1}B = B_Y$, we have $\operatorname{mult}_P g^*B_{X'} = \operatorname{mult}_P B_Y$ for any prime divisor P on Y which is not g-exceptional. Hence $g^*B_{X'} = B_Y$, and

$$K_Y + E + B_Y = g^*(K_{X'} + B_{X'}) = f^*(K_X + B).$$

Hence $(Y, E + B_Y)$ is lc. Construction-Proposition 4.2 follows. □

5. Proof of the main theorem

In this section we prove Theorem 1.2.

Proof of Theorem 1.2. If $\Gamma \subset \{0\}$, then (1) implies $B = B' = 0$, and the conclusion is (2). If $\dim X \leq 1$, then X is smooth, so $K_X + B$ is $\mathbb{R}$-Cartier, and (X, B) is lc as $B \in [0, 1]$. If there does not exist an $\mathbb{R}$-divisor $\Delta \geq 0$ on X such that $(X \ni x, \Delta)$ is an lc pair, then (2) does not hold for any B'. Hence any $\tau > 0$ satisfies the theorem if $\Gamma \subset \{0\}$, if $\dim X \leq 1$, or if no such Δ exists.

Thus we may assume that $\Gamma \not\subset \{0\}$, $\dim X \geq 2$, and there exists an $\mathbb{R}$-divisor $\Delta \geq 0$ on X such that $(X \ni x, \Delta)$ is an lc pair. We fix such Δ. Possibly shrinking X to a neighborhood of x, we may assume that X is affine and (X, Δ) is an lc pair.

Step 1. In this step we fix $f : Y \to X$, E, and ϵ.

We let $d := \dim X \geq 2$ and $\gamma := \min(\Gamma\backslash\{0\})$. Since (X, Δ) is an lc pair, by Construction-Proposition 4.2 there exists a projective birational morphism $f : Y \to X$ from a $\mathbb{Q}$-factorial variety Y satisfying the conclusion of Construction-Proposition 4.2. Let E be the reduced f-exceptional divisor.

Since $\{0, 1\} \subset [0, 1]$ and $\Gamma\backslash\{0\} \subset (0, +\infty)$ are finite, they are DCC sets. By Theorem 2.3, $\mathrm{LCT}_d(\{0, 1\}, \Gamma\backslash\{0\})$ satisfies the ACC. Thus there exists a real number $\epsilon \in (0, 1)$ such that

$$\mathrm{LCT}_d(\{0, 1\}, \Gamma\backslash\{0\}) \cap (1 - \epsilon, 1) = \emptyset, \tag{5.1}$$

since otherwise $\mathrm{LCT}_d(\{0, 1\}, \Gamma\backslash\{0\})$ contains a strictly increasing sequence converging to 1.

Step 2. In this step we define a subset $\mathcal{D}$ of a real vector space $\mathcal{V}$ and show that $\mathcal{D}$ is finite.

Let $q : Y' \to Y$ be a projective birational morphism from a smooth variety Y' such that, for $\pi := f \circ q$, $F := \pi^{-1}(x)$, with the reduced structure, is an snc divisor. Let $F_1, \dots, F_n$ be the irreducible components of F. Then F_k is a smooth projective variety and $\dim F_k = d - 1$ for any k. For any k, we let H_k be a very ample divisor on F_k.

Possibly shrinking X to a neighborhood of x, we may assume that X is affine and any π-exceptional prime divisor intersects F. Let $P_1, \dots, P_N$ be the π-exceptional prime divisors. Since $\dim F_k = d - 1$ and $\pi(F_k) = x$, any F_k is one of $P_1, \dots, P_N$. Since X is affine, possibly replacing K_X with $K_X + \mathrm{div}(\varphi)$ for some non-zero regular function φ on X, we may assume that $K_X \geq 0$. We let

$$\kappa_j := \mathrm{mult}_{P_j} K_{Y'} \in \mathbb{Z}$$

for any j.

Since Y' is smooth, any divisor D on Y' is Cartier. We let $D|_{F_k} \in N^1(F_k)_{\mathbb{R}}$ be the numerical class of the pullback of D under the inclusion $F_k \to Y'$, where D may contain F_k, and we extend $D \to D|_{F_k}$ linearly to $\mathbb{R}$-divisors D on Y'. We let

$$\deg_k(D) := (D|_{F_k}) \cdot H_k^{d-2}.$$

In particular, $\deg_k(P_j)$ may be positive or negative. We let

$$\mu_k := \sum_{j=1}^{N} (\kappa_j + 1)|\deg_k(P_j)| \quad \text{and} \quad \mu := \sum_{k=1}^{n} \mu_k.$$

We let

$$\mathcal{V} := \bigoplus_{k=1}^{n} N^1(F_k)_{\mathbb{R}} \quad \text{and} \quad v(D) := \left(D|_{F_k}\right)_{1 \leq k \leq n} \in \mathcal{V}$$

for any $\mathbb{R}$-divisor D on Y'. The map v is linear. Let $\mathcal{D} \subset \mathcal{V}$ be the set of $v(D)$, where D runs through the prime divisors on Y' such that $D \not\subset F$ and

$$\sum_{k=1}^{n} \deg_k(D) \leq \frac{2\mu}{\gamma}. \tag{5.2}$$

We show that $\mathcal{D}$ is finite. Let D be a prime divisor on Y' such that $D \not\subset F$ and (5.2) holds. For any k, since $D \not\supset F_k$, $D|_{F_k}$ is the class of a divisor $D_k \geq 0$ on F_k, so $\deg_k(D) = D_k \cdot H_k^{d-2}$ is an integer and

$$0 \leq \deg_k(D) \leq \frac{2\mu}{\gamma}.$$

For any integer $t \geq 0$, the divisors $R \geq 0$ on F_k with $R \cdot H_k^{d-2} = t$ are parametrized by the Chow variety of $(d-2)$-dimensional cycles on F_k of degree t with respect to $\mathcal{O}_{F_k}(H_k)$, which is projective [Kol96, Theorem I.3.21], hence has finitely many connected components. Two divisors $R, R' \geq 0$ on F_k parametrized by points of one connected component of the Chow variety are algebraically equivalent [Kol96, Definition II.4.1], hence numerically equivalent. Thus, for any k, $D|_{F_k}$ belongs to a finite subset of $N^1(F_k)_{\mathbb{R}}$ depending only on F_k, H_k, and $2\mu/\gamma$. Hence $\mathcal{D}$ is finite.

Step 3. In this step we define τ.

Let $J_q := \{j \mid P_j \text{ is } q\text{-exceptional}\}$. Any q-exceptional prime divisor is π-exceptional, hence equal to P_j for some $j \in J_q$. We let

$$\mathcal{V}_q := \mathrm{Span}_{\mathbb{R}}\{v(P_j) \mid j \in J_q\} \subset \mathcal{V}, \quad \mathcal{Q} := \mathcal{V}/\mathcal{V}_q,$$

and let $\varpi : \mathcal{V} \to \mathcal{Q}$ be the quotient map. We fix a norm $\|\cdot\|_{\mathcal{Q}}$ on $\mathcal{Q}$. Let $E_{Y'} := q_*^{-1}E$ and

$$\mathcal{S} := \left\{\varpi\Big(v(K_{Y'} + E_{Y'}) + \sum_{i=1}^{m} \beta_i w_i\Big) \;\Big|\; m \in \mathbb{Z},\ 0 \le m \le \frac{2\mu}{\gamma},\ \beta_i \in \Gamma\backslash\{0\},\ w_i \in \mathcal{D}\right\} \subset \mathcal{Q}.$$

Since Γ and $\mathcal{D}$ are finite, $\mathcal{S}$ is finite. We let

$$\eta := \min\{\|z\|_{\mathcal{Q}} \mid z \in \mathcal{S},\ z \neq 0\}$$

if $\mathcal{S} \not\subset \{0\}$, and $\eta := 1$ if $\mathcal{S} \subset \{0\}$. We let

$$\lambda := \max\left\{1, \frac{2\mu}{\gamma} \max_{w \in \mathcal{D}} \|\varpi(w)\|_{\mathcal{Q}}\right\}$$

if $\mathcal{D} \neq \emptyset$, and $\lambda := 1$ if $\mathcal{D} = \emptyset$. Finally, we let

$$\tau_1 := \frac{\eta}{2\lambda} \quad \text{and} \quad \tau := \min\left\{\frac{\gamma\epsilon}{2}, \tau_1\right\}.$$

Then $\tau > 0$, and τ depends only on $X \ni x$ and Γ.

Step 4. In this step we show that B has at most $2\mu/\gamma$ components and $v(\pi_*^{-1}B_i) \in \mathcal{D}$ for any component B_i of B.

Let B and B' be $\mathbb{R}$-divisors on X satisfying (1) and (2). Possibly shrinking X to a neighborhood of x, we may assume that any irreducible component of $\mathrm{Supp}\, B$ contains x and (X, B') is an lc pair. Since $0 \le B' \le B$, we may write

$$B = \sum_{i=1}^{m} b_i B_i \quad \text{and} \quad B' = \sum_{i=1}^{m} b_i' B_i,$$

where $B_1, \dots, B_m$ are distinct prime divisors, $b_i \in \Gamma\backslash\{0\}$, and $b_i' \ge 0$. Since $b_i \ge \gamma$ and

$$b_i - b_i' \le \|B - B'\| < \tau \le \frac{\gamma\epsilon}{2} < \frac{\gamma}{2} \quad \text{for any } i,$$

we have

$$b_i' > \frac{\gamma}{2} \quad \text{for any } i. \tag{5.3}$$

We write

$$\pi^*(K_X + B') = \pi_*^{-1}K_X + \sum_{i=1}^{m} b_i' \pi_*^{-1}B_i + \sum_{j=1}^{N} c_j' P_j,$$

where $c_j' := \mathrm{mult}_{P_j}\, \pi^*(K_X + B')$. Since $K_X + B' \ge 0$ is $\mathbb{R}$-Cartier, by Lemma 2.2, $c_j' \ge 0$. Since (X, B') is lc,

$$0 \le a(P_j, X, B') = 1 + \mathrm{mult}_{P_j}\left(K_{Y'} - \pi^*(K_X + B')\right) = 1 + \kappa_j - c_j'.$$

Thus

$$0 \le c_j' \le \kappa_j + 1 \quad \text{for any } j. \tag{5.4}$$

Fix k. Since $\pi(F_k) = x$, $\deg_k(\pi^*(K_X + B')) = 0$. Since $\pi_*^{-1}K_X \ge 0$ and $\pi_*^{-1}B_i \ge 0$ do not contain F_k, their restrictions to F_k are classes of divisors ≥ 0 on F_k, so $\deg_k(\pi_*^{-1}K_X) \ge 0$ and $\deg_k(\pi_*^{-1}B_i) \ge 0$. Taking $\deg_k$ of the decomposition of $\pi^*(K_X + B')$ above, since $\deg_k(\pi_*^{-1}K_X) \ge 0$ and by (5.4),

$$\sum_{i=1}^{m} b_i' \deg_k(\pi_*^{-1}B_i) \le -\sum_{j=1}^{N} c_j' \deg_k(P_j) \le \sum_{j=1}^{N} (\kappa_j + 1)|\deg_k(P_j)| = \mu_k. \tag{5.5}$$

Fix i. Since $x \in B_i$ and π is proper, $\pi_*^{-1}B_i \cap F \neq \emptyset$, so $\pi_*^{-1}B_i \cap F_k \neq \emptyset$ for some k. Since $\pi_*^{-1}B_i \not\supset F_k$, the restriction of $\pi_*^{-1}B_i$ to F_k is the class of a non-zero divisor ≥ 0 on F_k, so $\deg_k(\pi_*^{-1}B_i)$ is a positive integer. Thus

$$\sum_{k=1}^{n} \deg_k(\pi_*^{-1}B_i) \geq 1 \quad \text{for any } i. \tag{5.6}$$

By (5.6), (5.3), and the sum of (5.5) over k, for any i_0,

$$\frac{\gamma}{2}\max\Big\{m, \sum_{k=1}^{n}\deg_k(\pi_*^{-1}B_{i_0})\Big\} \leq \frac{\gamma}{2}\sum_{i=1}^{m}\sum_{k=1}^{n}\deg_k(\pi_*^{-1}B_i) \leq \sum_{i=1}^{m} b_i' \sum_{k=1}^{n}\deg_k(\pi_*^{-1}B_i) \leq \mu.$$

Thus

$$m \leq \frac{2\mu}{\gamma} \quad \text{and} \quad v(\pi_*^{-1}B_i) \in \mathcal{D} \text{ for any } i. \tag{5.7}$$

Step 5. In this step we show that there exist real numbers e_j, $j \in J_q$, such that

$$L_{Y'} := K_{Y'} + E_{Y'} + \pi_*^{-1}B + \sum_{j\in J_q} e_j P_j$$

satisfies $L_{Y'} \cdot C = 0$ for any curve $C \subset F$.

Let $B_Y' := f_*^{-1}B'$ and $B_Y := f_*^{-1}B$. By Construction-Proposition 4.2, possibly shrinking X to a neighborhood of x, we may assume that

$$K_Y + E + B_Y' = f^*(K_X + B')$$

and $(Y, E + B_Y')$ is lc. Thus $\pi^*(K_X + B') = q^*(K_Y + E + B_Y')$. Since $\pi(F_k) = x$ for any k, $v(\pi^*(K_X + B')) = 0$. Since

$$q_*\big(K_{Y'} + E_{Y'} + \pi_*^{-1}B'\big) = K_Y + E + B_Y',$$

the $\mathbb{R}$-divisor $q^*(K_Y + E + B_Y') - (K_{Y'} + E_{Y'} + \pi_*^{-1}B')$ is q-exceptional. Thus

$$\varpi\Big(v(K_{Y'} + E_{Y'}) + \sum_{i=1}^{m} b_i' v(\pi_*^{-1}B_i)\Big) = 0.$$

We let

$$z_B := \varpi\Big(v(K_{Y'} + E_{Y'}) + \sum_{i=1}^{m} b_i v(\pi_*^{-1}B_i)\Big) = \sum_{i=1}^{m}(b_i - b_i')\varpi\big(v(\pi_*^{-1}B_i)\big).$$

By (5.7), $z_B \in \mathcal{S}$. If $m = 0$, then $z_B = 0$. If $m \geq 1$, then $\mathcal{D} \neq \emptyset$, and since $0 \leq b_i - b_i' < \tau \leq \tau_1$,

$$\|z_B\|_{\mathcal{Q}} \leq \sum_{i=1}^{m}(b_i - b_i')\big\|\varpi\big(v(\pi_*^{-1}B_i)\big)\big\|_{\mathcal{Q}} \leq m\tau_1 \max_{w\in\mathcal{D}}\|\varpi(w)\|_{\mathcal{Q}} \leq \lambda\tau_1 = \frac{\eta}{2}.$$

Since $z_B \in \mathcal{S}$ and $\|z_B\|_{\mathcal{Q}} \leq \eta/2 < \eta$, $z_B = 0$ by the choice of η. Since

$$v(K_{Y'} + E_{Y'} + \pi_*^{-1}B) = v(K_{Y'} + E_{Y'}) + \sum_{i=1}^{m} b_i v(\pi_*^{-1}B_i),$$

we have $v(K_{Y'} + E_{Y'} + \pi_*^{-1}B) \in \mathcal{V}_q$, so there exist real numbers e_j, $j \in J_q$, such that

$$v(L_{Y'}) = 0.$$

Since any curve $C \subset F$ is contained in F_k for some k, $L_{Y'} \cdot C = 0$.

Step 6. In this step we show that $K_Y + E + B_Y$ is numerically trivial over X.

By Lemma 2.5, possibly shrinking X to a neighborhood of x, we may assume that $L_{Y'}$ is numerically trivial over X. Since Y is $\mathbb{Q}$-factorial, $K_Y + E + B_Y$ is $\mathbb{R}$-Cartier. Since

$q_*L_{Y'} = K_Y + E + B_Y$, the $\mathbb{R}$-divisor $L_{Y'} - q^*(K_Y + E + B_Y)$ is q-exceptional. It is numerically trivial over Y as any curve contracted by q is contracted by π. By the negativity lemma,

$$L_{Y'} = q^*(K_Y + E + B_Y).$$

Since $q^*(K_Y + E + B_Y) = L_{Y'}$ is numerically trivial over X and q is surjective, $K_Y + E + B_Y$ is numerically trivial over X by the projection formula.

Step 7. In this step we show that $(Y, E + B_Y)$ is lc.

By **Step 5**, $(Y, E + B'_Y)$ is lc. For any i, since $b_i \geq \gamma$ and $b_i - b'_i < \tau \leq \gamma\epsilon/2$,

$$b'_i - \Big(1 - \frac{\epsilon}{2}\Big)b_i = \frac{\epsilon}{2}b_i - (b_i - b'_i) > \frac{\epsilon}{2}\gamma - \frac{\gamma\epsilon}{2} = 0.$$

Since $(Y, E + B'_Y)$ is lc, Y is $\mathbb{Q}$-factorial, and $0 \leq (1 - \epsilon/2)B_Y \leq B'_Y$, the pairs (Y, E) and $(Y, E + (1 - \epsilon/2)B_Y)$ are lc.

Since (Y, E) is an lc pair of dimension d with $E \in \{0, 1\}$, and $B_Y \geq 0$ is $\mathbb{R}$-Cartier with $B_Y \in \Gamma\backslash\{0\}$,

$$\mathrm{lct}(Y, E; B_Y) \in \mathrm{LCT}_d(\{0, 1\}, \Gamma\backslash\{0\}) \quad \text{and} \quad \mathrm{lct}(Y, E; B_Y) \geq 1 - \frac{\epsilon}{2} > 1 - \epsilon.$$

By (5.1), $\mathrm{lct}(Y, E; B_Y) \geq 1$, so $(Y, E + B_Y)$ is lc.

Step 8. In this step we conclude the proof.

Since $f_*(E + B_Y) = B$, $(Y, E + B_Y)$ is an lc pair by **Step 7**, and $K_Y + E + B_Y$ is numerically trivial over X by **Step 6**, Lemma 2.4 implies that $K_X + B$ is $\mathbb{R}$-Cartier and (X, B) is lc. Since X was replaced only by open neighborhoods of x, $K_X + B$ is $\mathbb{R}$-Cartier near x and $(X \ni x, B)$ is lc. □

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